

ELEC 402 Project 3: Dot-product-based DNN Accelerator

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Design Specs

Post-Synthesis Area: 15803.478 μm^2

	Setup Slack (ps)	Hold Slack(ps)	Area(μm^2)
Post-Layout	447	23	27669.510

Layout Pictures

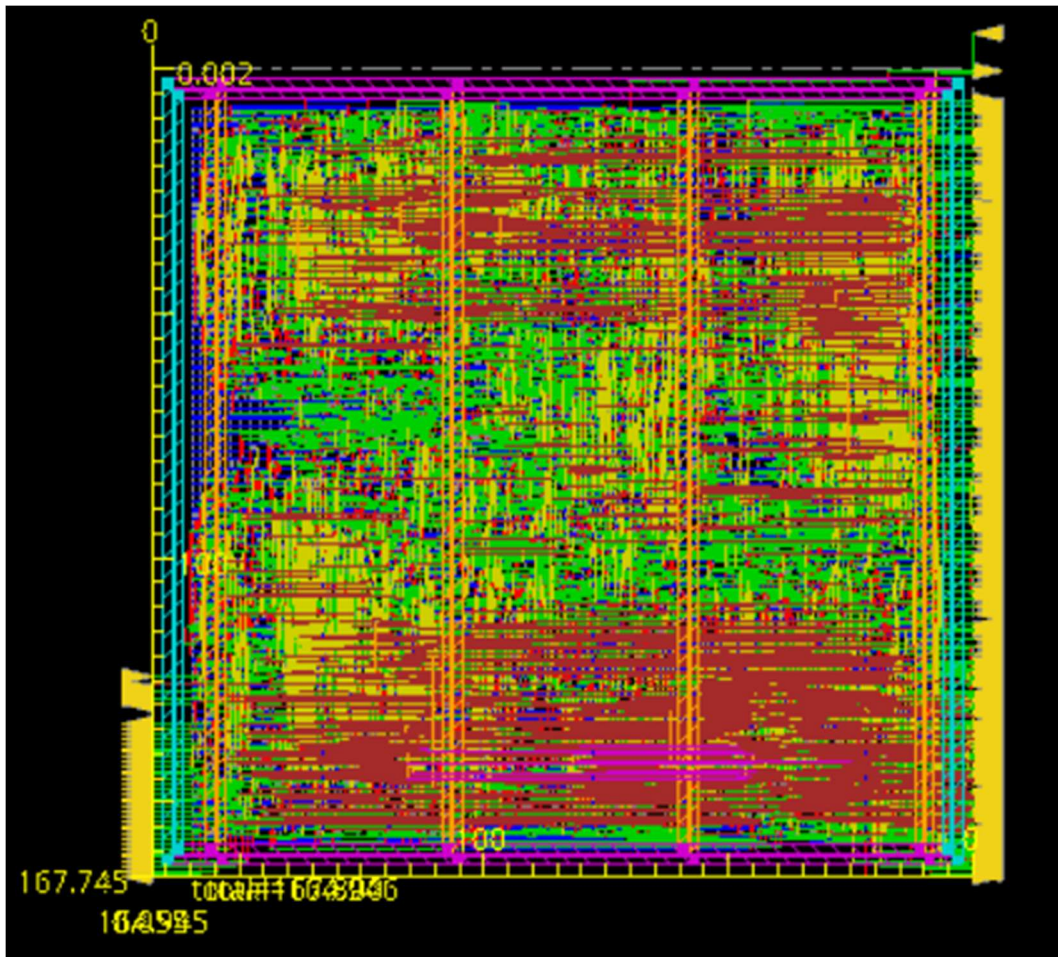


Figure 1: Layout with edge dimensions marked

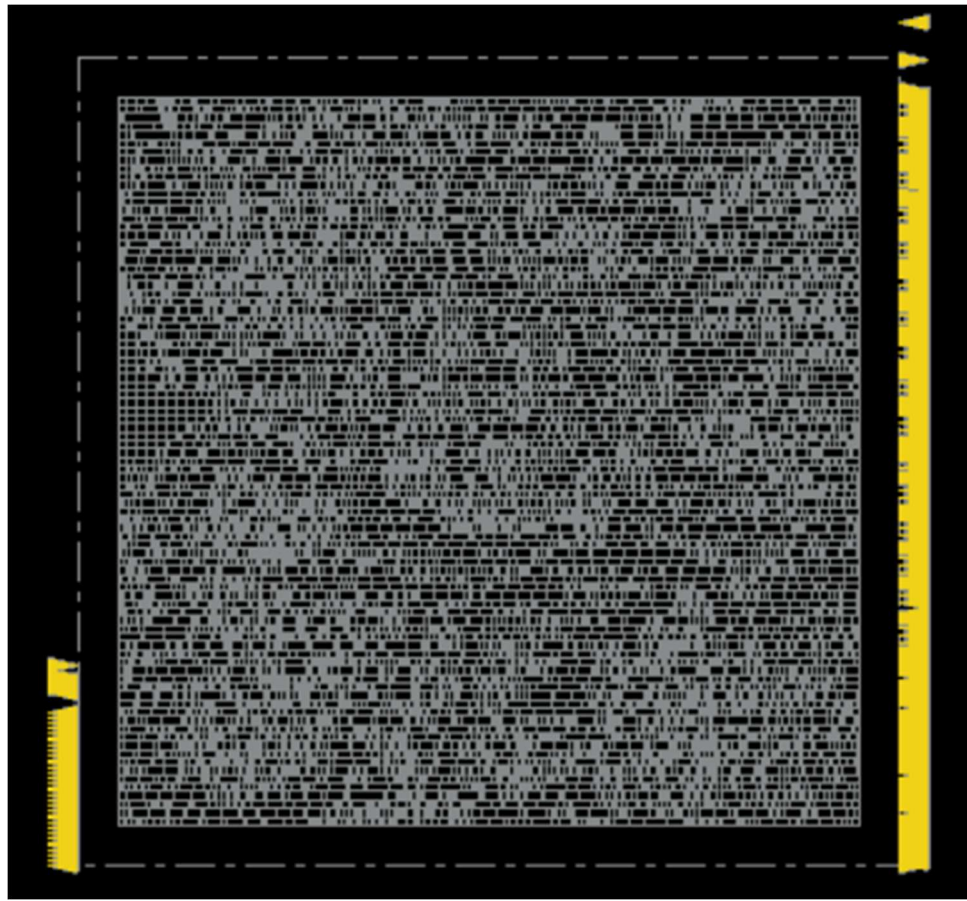


Figure 2: Layout with routing layers turned off

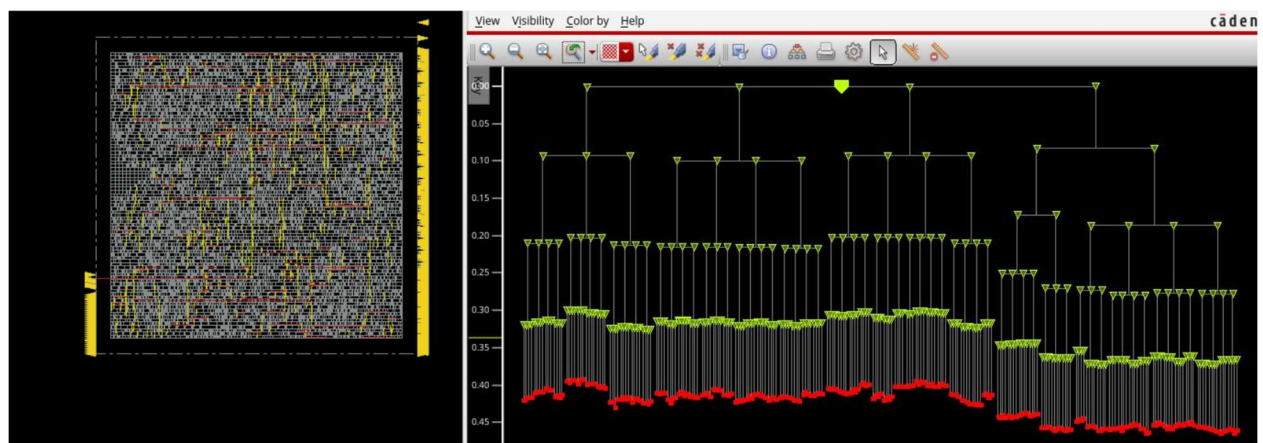


Figure 3: Layout with clock tree marked and ccopt on the side

RTL Simulation

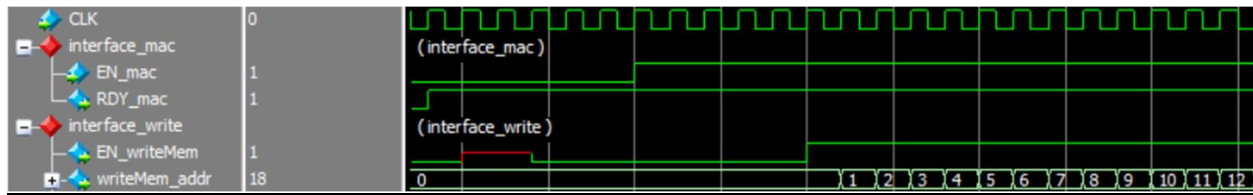


Figure 4: Waveforms for the control signals during MAC operation



Figure 5: Waveforms for the control signals when memory is full

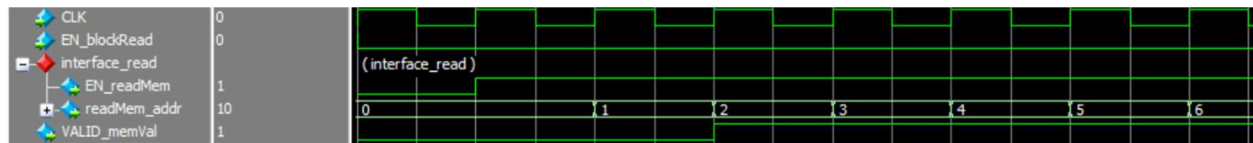


Figure 6: Waveforms during read operations on the memories

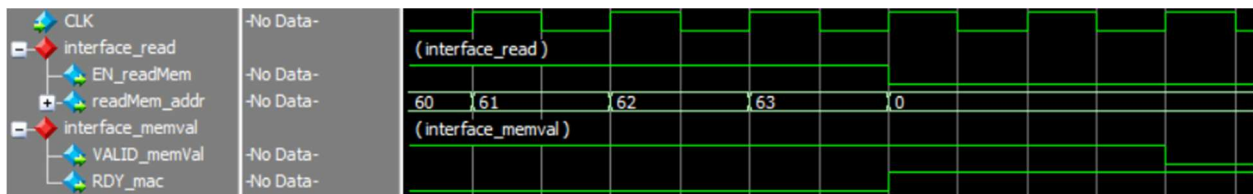


Figure 7: Waveforms at the end of read operations

PNR Layout Simulation

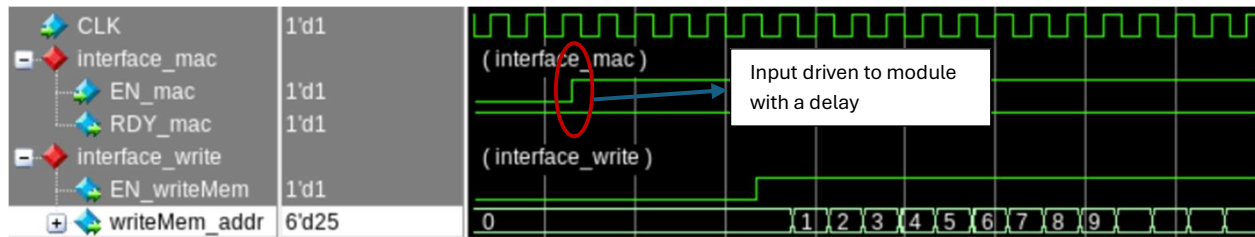


Figure 8: Waveforms for the control signals during MAC operation (PNR)

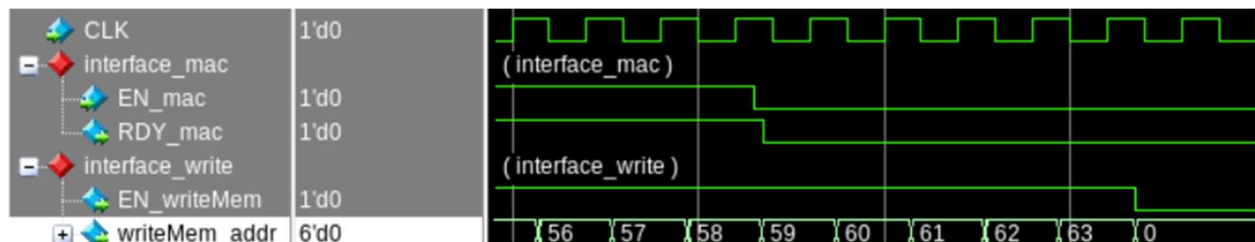


Figure 9: Waveforms for the control signals when memory is full (PNR)

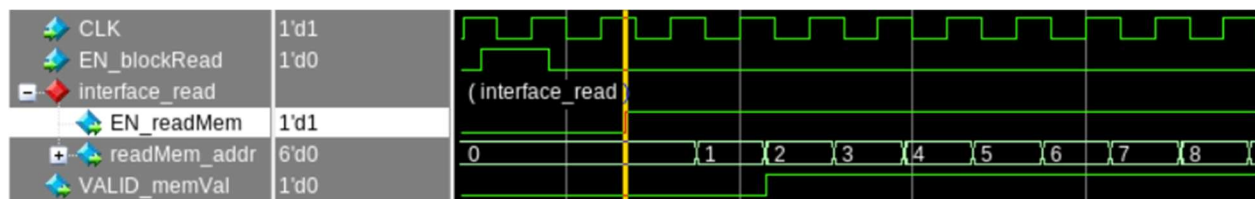


Figure 10: Waveforms during read operations on the memories (PNR)

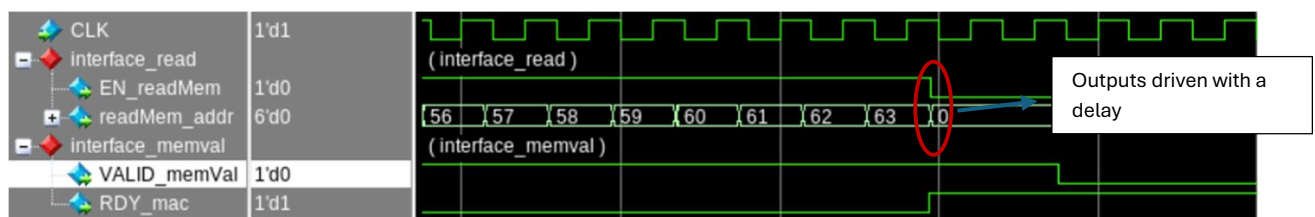


Figure 11: Waveforms at the end of read operations (PNR)

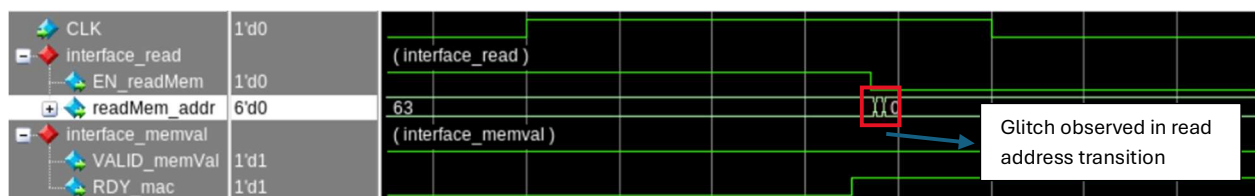


Figure 12: Zoomed in picture of the read interface to show glitches present in design


```
# =====  
#   Test Summary  
# =====  
#   Total Iterations:    8  
#   Total Writes:       512  
#   Total Reads:        512  
#   Expected Total:     512  
#   Verification Errors:0  
# =====
```

Figure 13: Output produced by testbench to check how many operations are done correctly (8 Read/Write Iterations and 0 errors found)

The following changes were made to the testbench to comply with PNR netlist simulation:

- An IO clock with a 500ps delay was introduced to ensure readMem_val was fed into the module with the correct timing. This clock is attached at the read port of the memory.
- Rest other module inputs were fed from the testbench with a 500ps delay.

NOTE: All relevant files have been added along with this report for reference. This includes files for part 3, 4, 6, 7, 8